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Foreword

This Technical Specification (TS) has been produced by ETSI Technical Committee Smart Machine-to-Machine communications (SmartM2M).

The present document is part 1 of a multi-part deliverable covering SmartM2M; Extension to SAREF, as identified below:

Part 1: "Energy Domain";

Part 2: "Environment Domain";

Part 3: "Building Domain";

Part 4: "Smart Cities Domain";

Part 5: "Industry and Manufacturing Domains";

Part 6: "Smart Agriculture and Food Chain Domain";

Part 7: "Automotive Domain";

Part 8: "eHealth/Ageing-well Domain";

Part 9: "Wearables Domain";

Part 10: "Water Domain".

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

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1 Scope

The present document presents SAREF4ENER, the SAREF extension for EEBus and Energy@Home in the energy domain.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

Referenced documents which are not found to be publicly available in the expected location might be found at https://docbox.etsi.org/Reference/.

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The following referenced documents are necessary for the application of the present document.

[1] EEBus SPINE.

NOTE: Available at https://www.eebus.org/en/specifications/.

[2] ETSI TS 103 264 (V3.1.1): "SmartM2M; Smart Applications; Reference Ontology and oneM2M Mapping".

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long term validity.

The following referenced documents are not necessary for the application of the present document but they assist the user with regard to a particular subject area.

[i.1] TNO, EEBus, Energy@Home: "SAREF4EE".

NOTE: Available at https://w3id.org/saref4ee.

[i.2] Energy@home Data Model, v2.1, October 2015.

NOTE: Available at http://www.energy-home.it/Documents/Technical%20Specifications/E@h_data_model_v2.1.pdf.

[i.3] IEC TR 62746-2:2015: "Systems interface between customer energy management system and the power management system - Part 2: Use cases and requirements".

NOTE: Available at https://webstore.iec.ch/publication/22279.

[i.4] ETSI TR 103 411: "SmartM2M Smart Appliances SAREF extension investigation".

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the following terms apply:

ontology: formal specification of a conceptualization, used to explicit capture the semantics of a certain reality

3.2 Symbols

Void.

3.2 Abbreviations

For the purposes of the present document, the following abbreviations apply:

CEM	Customer Energy Manager
E@H	Energy@Home association
EEBus	EEBus initiative
OM	Ontology of units of Measure
OWL	Web Ontology Language
SAREF	Smart Applications REFerence ontology
TNO	Netherlands Organization for Applied Scientific Research
TR	Technical Report
TS	Technical Specification
UML	Unified Modeling Language
XSD	W3C XML Schema Definition

4 SAREF4ENER ontology and semantics

4.1 Introduction and overview

The present document is a technical specification of SAREF4ENER, an extension of SAREF [2] that was created in collaboration with Energy@Home (http://www.energy-home.it) and EEBus (http://www.eebus.org/en), the major Italy- and Germany-based industry associations, to enable the interconnection of their (different) data models. The Energy@Home association, abbreviated in the rest of the document as E@H. E@H aims at developing and promoting technologies and services for energy efficiency in smart homes, based upon the interaction between user devices and the energy infrastructure. The E@H data model is described in [i.2]. EEBus is an important initiative in the area of the Internet of Things, which has its roots in the sector of smart and renewable energy. EEBus developed a standardized and consensus-oriented smart grid and smart home networking concept. The EEBus data model is described in [1]. SAREF4ENER is meant to enable the (currently missing) interoperability among various proprietary solutions developed by different consortia in the smart home domain. By using SAREF4ENER, smart appliances from manufacturers that support the EEBus or E@H data models will easily communicate with each other using any energy management system at home or in the cloud.

Towards this aim, SAREF4ENER should be used to annotate (or generate) a neutral (protocol-independent) set of messages to be directly adopted by the various manufacturers, or mapped to their domain specific protocols of choice.

SAREF4ENER is an OWL-DL ontology that extends SAREF with 63 classes, 17 object properties and 40 data type properties. SAREF4ENER focuses on demand response scenarios, in which customers can offer flexibility to the Smart Grid to manage their smart home devices by means of a Customer Energy Manager (CEM). The CEM is a logical function for optimizing energy consumption and/or production that can reside either in the home gateway or in the cloud. Moreover, the Smart Grid can influence the quantity or patterns of use of the energy consumed by customers when energy-supply systems are constrained, e.g. during peak hours. These scenarios involve the following use cases:

- **Use case 1:** configuration of devices that want to connect to each other in the home network, for example, to register a new dishwasher to the list of devices managed by the CEM;
- **Use case 2:** smart energy management/ (re-)scheduling appliances in certain modes and preferred times using power profiles to optimize energy efficiency and accommodate the customer's preferences;
- **Use case 3:** monitoring and control of the start and status of the appliances;
- **Use case 4:** reaction to special requests from the Smart Grid, for example, incentives to consume more or less depending on current energy availability, or emergency situations that require temporary reduction of the power consumption.

These use cases are associated with the user stories described in [i.3], which include, among others, the following examples:

- User wants to do basic settings of his/her devices;
- User wants to know when the washing machine has finished working;
- User wants the washing done by 5:00 p.m. with least electrical power costs;
- User likes to limit his/her own energy consumption up to a defined limit;
- User allows the CEM to reduce the energy consumption of his/her freezer in a defined range for a specific time, if the grid recognizes (severe) stability issues;
- Grid related emergency situations (blackout prevention).

The prefixes and namespaces used in SAREF4ENER and in the present document are listed in Table 1.

Table 1: Prefixes and namespaces used within the SAREF4ENER ontology

Prefix	Namespace
s4ener	https://saref.etsi.org/saref4ener/
saref	https://saref.etsi.org/core/
dcterms	http://purl.org/dc/terms/
owl	http://www.w3.org/2002/07/owl#
rdf	http://www.w3.org/1999/02/22-rdf-syntax-ns#
rdfs	http://www.w3.org/2000/01/rdf-schema#
om	http://www.wurvoc.org/vocabularies/om-1.8/
xsd	http://www.w3.org/2001/XMLSchema#

4.2 SAREF4ENER

4.2.1 General Overview

An overview of the SAREF4ENER ontology is provided in Figure 1, where rectangles containing an orange circle are used to denote classes created in SAREF4ENER, while rectangles containing a faded orange circle denote classes reused from other ontologies, such as SAREF. For all the entities described in the present document, it is indicated whether they are defined in the SAREF4ENER extension or elsewhere by the prefix included before their identifier, i.e. if the element is defined in SAREF4ENER the prefix is `s4ener:`, while if the element is reused from another ontology it is indicated by a prefix according to Table 1.

Arrows with white triangles on top represent the `rdfs:subClassOf` relation between two classes. The origin of the arrow is the class to be declared as subclass of the class at the destination of the arrow.

Directed arrows are used to represent properties between classes.

Rectangles that contain a list of values between square brackets denote an enumeration of individuals.

Note that Figure 1 aims at showing a global overview of the main classes of SAREF4ENER and their mutual relations. More details on the different parts of Figure 1 are provided in clause 4.2.2 to clause 4.2.6.

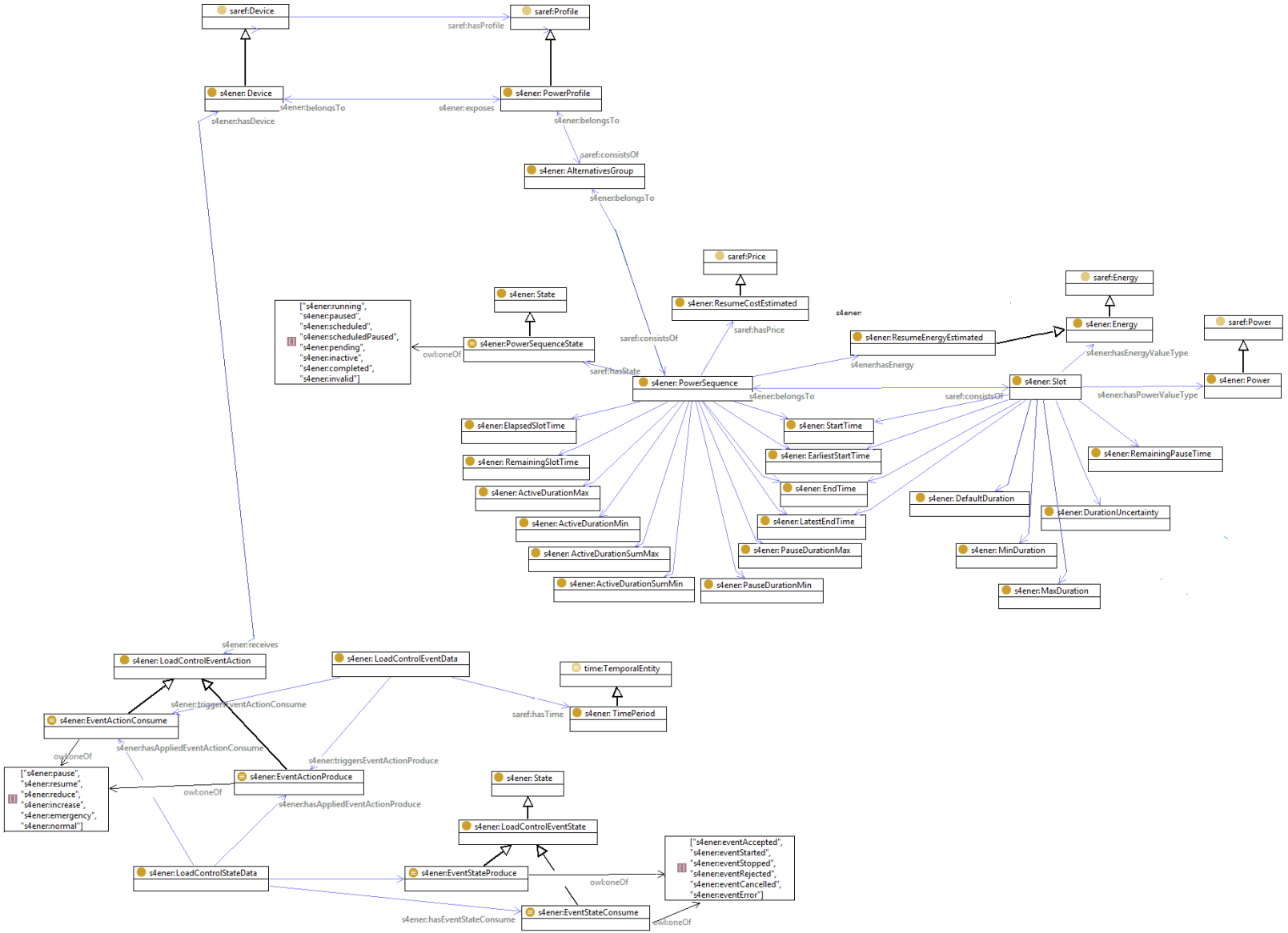


Figure 1: SAREF4ENER overview

Figure 2 shows the hierarchy of classes and properties defined in SAREF4ENER.

Orange circles represent classes of SAREF4ENER, while faded orange circles represent classes that are reused from other ontologies. Object properties - which are properties between two classes - are denoted by blue rectangles, while datatype properties - which are properties between a class and a data type, such as `xsd:string` or `xsd:dateTime` - are denoted by green rectangles. Faded blue and green rectangles denote object properties and datatype properties that are reused from other ontologies.



Figure 2: SAREF4ENER class and property hierarchy

4.2.2 Device

A `s4ener:Device` is a subclass of a `saref:Device`, i.e. it inherits the properties of the more general saref:Device and extends it with additional properties that are specific for SAREF4ENER. The `s4ener:Device` class is shown in Figure 3.

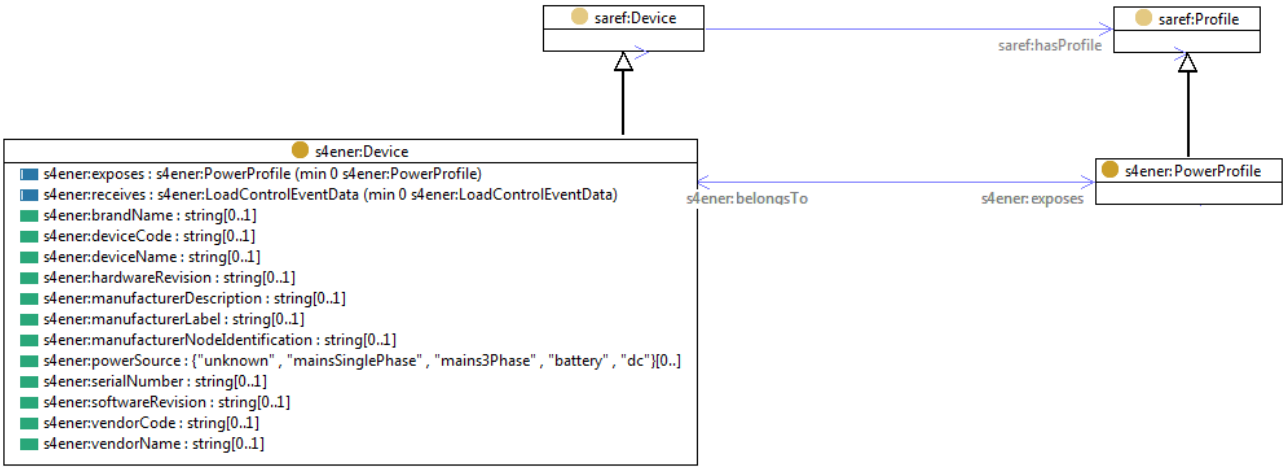


Figure 3: Device

Table 2 summarizes the properties that characterize a `s4ener:Device`.

Table 2: Properties of a Device

Property	Definition
s4ener:exposes **min** 0 s4ener:PowerProfile	A relationship between a device and its power profile.
s4ener:receives **min** 0 s4ener:LoadControlEventData	A relationship between a device (e.g. an appliance or a smart meter) and a load control event.
s4ener:brandName **max** 1 xsd:string	The name of the brand of a device. Useful where the name of the brand and the vendor differs.
s4ener: deviceCode **max** 1 xsd:string	Device code for the device as defined by the manufacturer.
s4ener: s4ee:deviceName **max** 1 xsd:string	Name of the device as defined by the manufacturer.
s4ener: hardwareRevision **max** 1 xsd:string	Hardware revision of the device as defined by the manufacturer.
s4ener:manufacturerDescription **max** 1 xsd:string	A description for the device as defined by the manufacturer.
s4ener:manufacturerLabel **max** 1 xsd:string	A short label of the device as defined by the manufacturer.
s4ener:manufacturerNodeIdentification **max** 1 xsd:string	A node identification for the device as defined by the manufacturer. This could be used for the identification of a device, even if it was removed from the network and re-joined later with changed node address.
s4ener:powerSource **min** 0 xsd:string	The power source of a device. Possible values are {"unknown", "mainsSinglePhase", "mains3Phase", "battery", "dc"}.
s4ener:serialNumber **max** 1 xsd:string	Serial number of a device as defined by the manufacturer. Usually the same as printed on the case.
s4ener:softwareRevision **max** 1 xsd:string	Software revision of a device as defined by the manufacturer.
s4ener:vendorCode **max** 1 xsd:string	Code for the vendor of the device as defined by the manufacturer.
s4ener:vendorName **max** 1 xsd:string	Name of the vendor of the device as defined by the manufacturer.

4.2.3 Power Profile and Alternatives Group

This clause presents the classes of interest for smart energy management. These classes are used to schedule devices in certain modes and preferred times using power profiles to optimize energy efficiency and accommodate the customer's preferences (i.e. use case 2). These classes are `s4ener:PowerProfile`, `s4ener:Alternative`, `s4ener:PowerSequence` and `s4ener:Slot`, which are shown in Figure 4.

A `s4ener:PowerProfile` is a subclass of a `saref:Profile`, i.e. it inherits the properties of the more general `saref:Profile` extending it with additional properties that are specific for SAREF4ENER. The `s4ener:PowerProfile` is used by a `s4ener:Device` to expose the power sequences that are potentially relevant for the CEM. A `s4ener:Device` can expose a `s4ener:PowerProfile`, which consists of one or more alternative plans (`s4ener:AlternativesGroup` class). A `s4ener:AlternativesGroup` consists of one or more power sequences (`s4ener:PowerSequence` class), and a `s4ener:PowerSequence` consists of one or more slots (`s4ener:Slot` class). Inversely, a `s4ener:Slot` belongs to only and exactly one `s4ener:PowerSequence`, which, in turn, belongs to only and exactly one `s4ener:AlternativesGroup`, which, in turn, belongs to only and exactly one `s4ener:PowerProfile`. A `s4ener:PowerProfile` belongs to only and exactly one `s4ener:Device`.

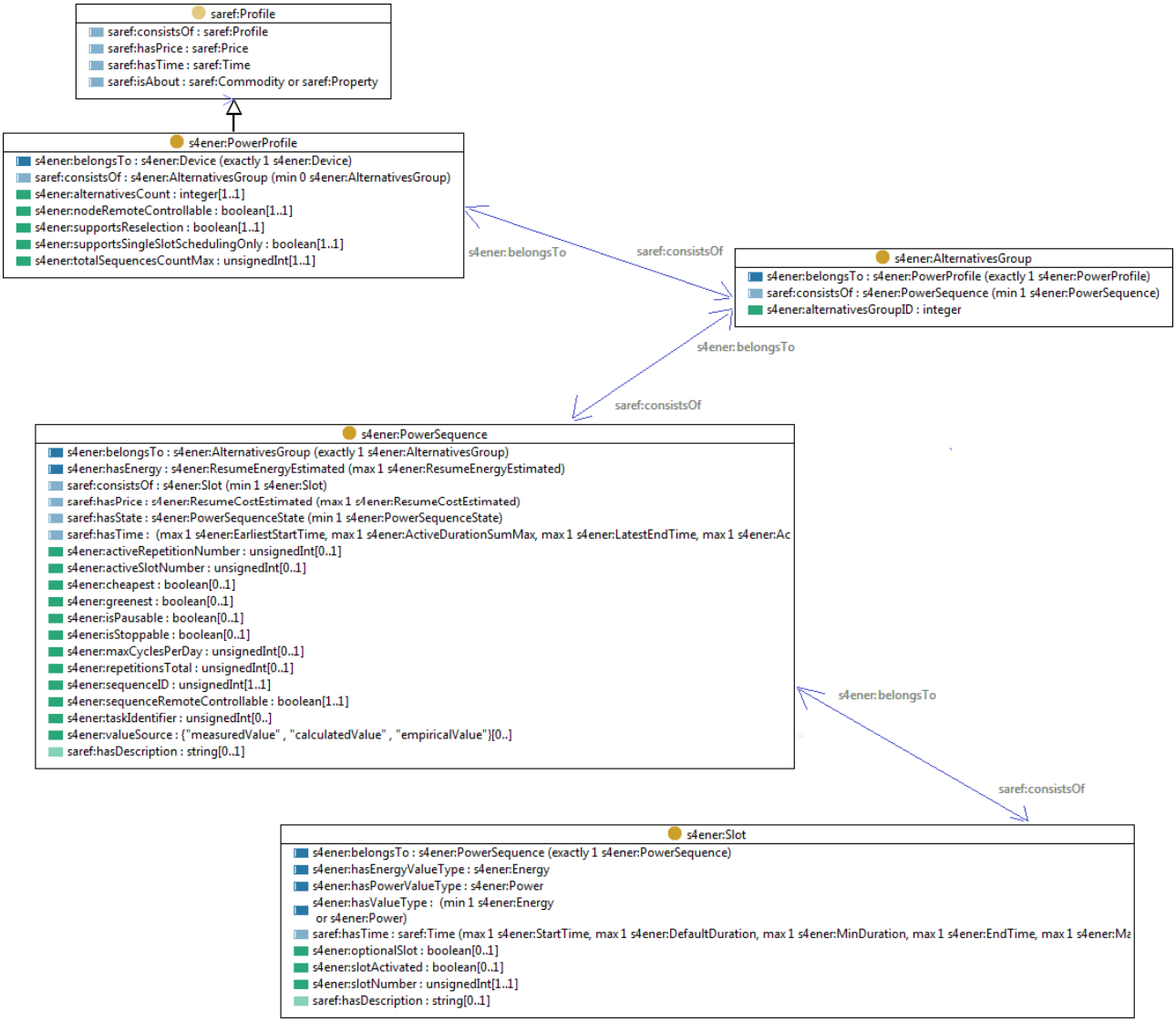


Figure 4: Power Profile and Alternatives Group

Table 3 summarizes the properties that characterize a `s4ener:PowerProfile` and an `s4ener:AternativesGroup`.

Table 3: Properties of a Power Profile and an AlternativesGroup

Property	Definition
s4ener:alternativesGroupID **exactly** 1 xsd:unsignedInt	The endpoint-wide unique identifier for the alternatives group instances provided by a power profile.
s4ener:alternativesCount **exactly** 1 xsd:integer	Number of "alternatives" groups provided by a power profile.
s4ener:nodeRemoteControllable **exactly** 1 xsd:boolean	Whether the device is configured for remote control by the CEM. This refers to the selection chosen by the user on the remote control feature of the device.
s4ener:supportsReselection **exactly** 1 xsd:boolean	Whether the device restricts the number of sequence re-selections by the CEM. If set to TRUE, there is no restriction, i.e. within a given alternative the CEM may first choose one sequence, alter the selection by configuring another sequence later on, then alter the selection again, etc. If set to FALSE, the device permits the CEM to select a sequence of an alternative only one time.
s4ener:supportsSingleSlotSchedulingOnly **exactly** 1 xsd:boolean	Whether the device permits the modification of more than one slot per configuration command. If set to TRUE the device does NOT permit this modification.
s4ener:totalSequencesCountMax **exactly** 1 xsd:unsignedInt	Total number of sequences supported by the device, i.e. the sum of all power sequences across all alternatives.

4.2.4 Power Sequence

The `s4ener:AlternativesGroup` described in clause 4.2.3 consists of one or more power sequences (`s4ener:PowerSequence` class) and, inversely, a `s4ener:PowerSequence` belongs to only and exactly one `s4ener:AlternativesGroup`. Figure 5 shows the details of the `s4ener:PowerSequence` class.

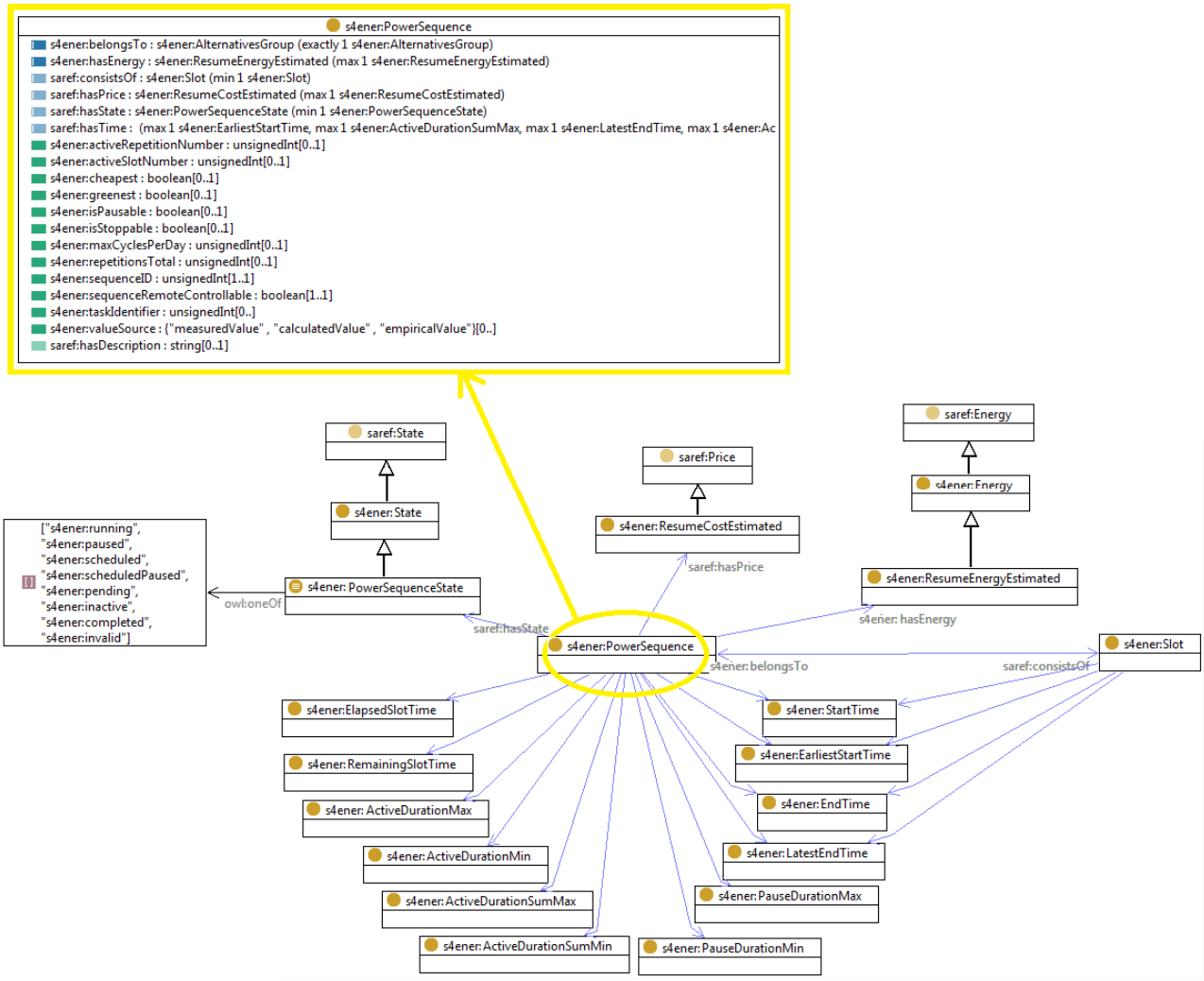


Figure 5: Power Sequence

Table 4 summarizes the properties that characterize a `s4ener:PowerSequence`.

Table 4: Properties of the PowerSequence

Property	Definition
s4ener:sequenceId **exactly** 1 xsd:unsignedInt	An endpoint-wide unique sequence identifier.
saref:hasDescription **max** 1 xsd:string	Textual description for the power sequence.
s4ener:isStoppable **max** 1 xsd:boolean	If the power sequence is stoppable by the CEM, this element is TRUE. Otherwise it SHALL be omitted.
s4eef:isPausable **max** 1 xsd:boolean	If the power sequence is pausable by the CEM, this element is TRUE. Otherwise it SHALL be omitted.
s4ener:taskIdentifier **min** 0 xsd:unsignedInt	Used by a device that wants to uniquely identify reoccurring types of power sequences. For example, specific types of washing cycles with specific parameters SHOULD have the same s4ener:taskIdentifier value every time they are offered using power sequences.
s4ener:activeRepetitionNumber **max** 1 xsd:unsignedInt	The current repetition of the sequence of slots. SHALL be present if s4ener:repetitionsTotal is present and has a value > 1. Otherwise, it SHALL be absent.
s4ener:activeSlotNumber **max** 1 xsd:unsignedInt	If s4ener:PowerSequenceState is set to "running" or "paused" this element SHALL contain the currently active slot. Otherwise it SHALL be omitted.
s4ener:cheapest **max** 1 xsd:boolean	If present and set to TRUE, the CEM SHALL try to apply a configuration that minimizes the user's energy bill for this power sequence. Absence of this element is equal to the presence with value FALSE.

Property	Definition
s4ener:greenest **max** 1 xsd:boolean	If present and set to TRUE, the CEM SHALL try to optimize the configuration towards the maximum availability of renewable energy. Absence of this element is equal to the presence with value FALSE.
s4ener:maxCyclesPerDay **max** 1 xsd:unsignedInt	The maximum amount of starts that the device allows per day.
s4ener:repetitionsTotal **max** 1 xsd:unsignedInt	If a power sequence repeats its sequence of slots, the element is present and contains the total number of repetitions. Absence of the element is equal to a presence with a value of 0 (zero). SHALL be absent if the value is 1.
s4ener:sequenceRemoteControllable **exactly** 1 xsd:boolean	Whether the sequence is modifiable (if value is TRUE) or not (if value is FALSE). Modifiability is required to configure power sequences and slots. It is also required to change a power sequence state.
s4ener:valueSource **min** 0 {"measuredValue", "calculatedValue", "empiricalValue"}	The source (origin/foundation) of the measurement forecasted values for this power sequence. If absent, the source is undefined.
s4ener:hasEnergy **max** 1 s4ener:ResumeEnergyEstimated	The additional energy the device will consume before resuming its normal operation (after a pause). This is only an estimated value which will not be added to the value stated in any slot value information.
saref:hasPrice **max** 1 s4ener:ResumeCostEstimated	The additional costs for the resumption of a device to its normal operation (after a pause).
saref:hasState **min** 1 s4ener:PowerSequenceState	The current state of the power sequence. It can assume one of the following values: 'running', 'paused', 'scheduled', 'scheduled paused', 'pending', 'inactive', 'completed' or 'invalid'.
saref:hasTime **max** 1 s4ener:ActiveDurationMax	The active maximum duration the power sequence can run without interruption.
saref:hasTime **max** 1 s4ener:ActiveDurationMin	The active minimum duration the power sequence can run without interruption.
saref:hasTime **max** 1 s4ener:ActiveDurationSumMax	The active maximum duration the power sequence can run in total (summation of all active times).
saref:hasTime **max** 1 s4ener:ActiveDurationSumMin	The active minimum duration the power sequence runs in total (summation of all active times).
saref:hasTime **min** 1 s4ener:StartTime	The start time of the power sequence. SHALL be present.
saref:hasTime **max** 1 s4ener:EarliestStartTime	SHALL state the earliest possible start time for the whole power sequence.
saref:hasTime **max** 1 s4ener:EndTime	The end time of the power sequence. If the value is available, it SHALL be denoted here. Otherwise the element SHALL be omitted.
saref:hasTime **max** 1 s4ener:LatestEndTime	The latest possible end time for the whole power sequence.
saref:hasTime **max** 1 s4ener:ElapsedSlotTime	If the power sequence state is set to 'running' or 'paused' AND the slot is determined, this element CAN contain the time the slot has already been in 'running' state (this also means the value remains constant during a 'paused' state). Otherwise it SHALL be omitted.
saref:hasTime **max** 1 s4ener:RemainingSlotTime	If the power sequence state is set to 'running' or 'paused' AND the slot is determined, this element SHALL contain the time the slot still needs to be in 'running' state (this also means the value remains constant during a 'paused' state). Otherwise it SHALL be omitted.
saref:hasTime **max** 1 s4ener:PauseDurationMax	The maximum duration the power sequence can pause after the end of an activity.
saref:hasTime **max** 1 s4ener:PauseDurationMin	The minimum duration the power sequence can pause after the end of an activity.

4.2.5 Slot

The `s4ener:PowerSequence` described in clause 4.2.4 consists of one or more slots (`s4ener:Slot` class) and, inversely, a `s4ener:Slot belongs` to only and exactly one `s4ener:PowerSequence`. Figure 6 shows the details of the s4ener:Slot class.

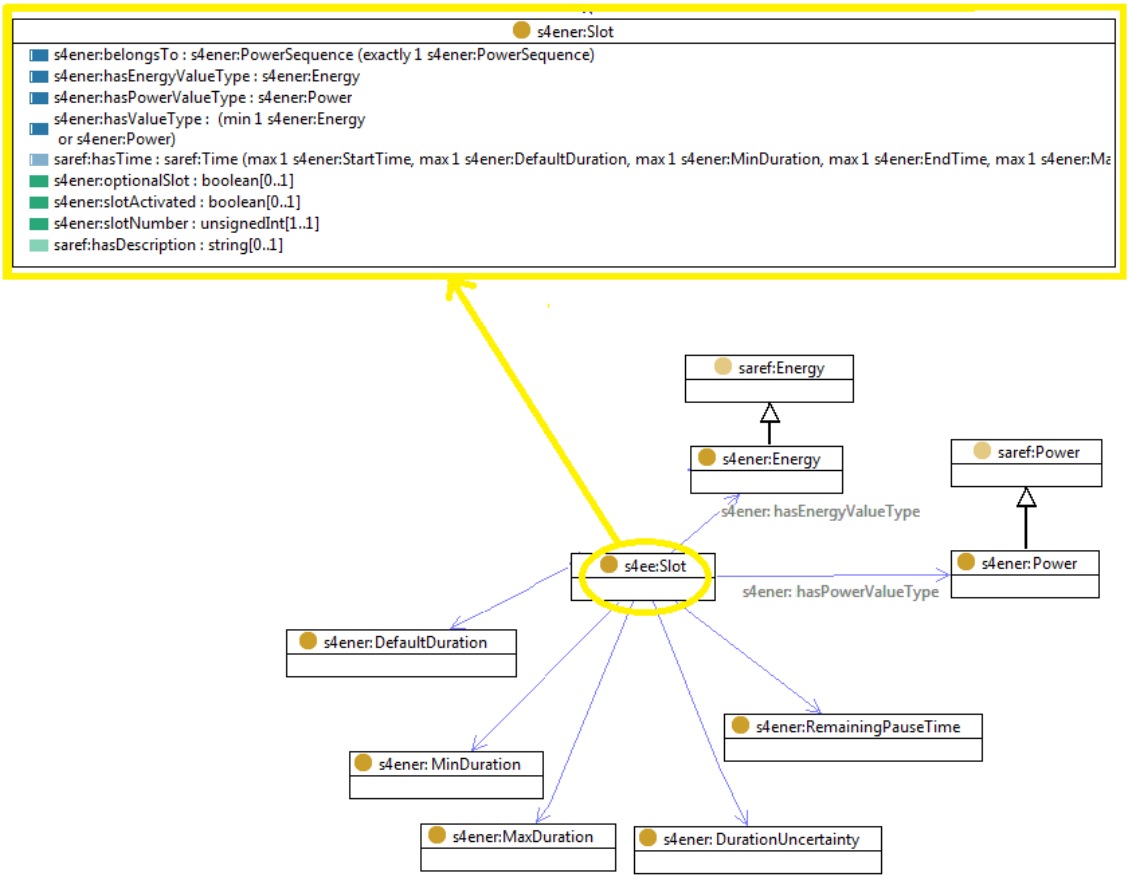


Figure 6: Slot

Table 5 summarizes the properties that characterize a `s4ener:Slot`.

Table 5: Properties of a Slot

Property	Definition
s4ener:slotNumber **exactly** 1	A power sequence Id-wide unique slot identifier.
saref:hasDescription **max** 1	Textual description for the slot.
s4ener:optionalSlot **max** 1	It is set to TRUE if the slot can be omitted, otherwise the element SHALL be omitted or set to FALSE (see note 1).
s4ener:slotActivated **max** 1	If the slot is optional, i.e. s4ener:optionalSlot is set to TRUE, this element reflects the current status of the slot (TRUE = the slot will be executed, FALSE = the slot will not be executed). If the slot is not optional, this element SHALL be absent.
s4ener:hasValueType **min** 1 (s4ener:Energy **or** s4ener:Power)	The type of energy or power (subclasses of saref:Energy and saref:Power). The energy can be of type s4ener:EnergyMin, s4ener:EnergyMax, s4ener:EnergyExpected, s4ener:EnergyStandardDeviation or s4ener:EnergySkewness. The power can be of type s4ener:PowerMin, s4ener:PowerMax, s4ener:PowerExpected, s4ener:PowerStandardDeviation or s4ener: Power Skewness.
saref:hasTime **max** 1 s4ener:DefaultDuration	The duration of the slot (in case of 'determined slot'). If the slot has a configurable length, this element SHALL reflect the currently configured length.
saref:hasTime **max** 1 s4ener:MaxDuration	The maximum supported configuration (if the slot has a configurable duration).
saref:hasTime **max** 1 s4ener:MinDuration	The minimum supported configuration (if the slot has a configurable duration) (see note 3).
saref:hasTime **max** 1 s4ener:DurationUncertainty	The uncertainty of the duration given in the s4ener:Duration class.
saref:hasTime **max** 1 s4ener:StartTime	The start time of the slot. SHALL be present.
saref:hasTime **max** 1 s4ener:EarliestStartTime	SHALL state the earliest possible start time for the slot.
saref:hasTime **max** 1 s4ener:EndTime	The end time of the slot. The following equation SHALL apply: EndTime - StartTime = DefaultDuration.
saref:hasTime **max** 1 s4ener:LatestEndTime	The latest possible end time for the slot.
saref:hasTime **max** 1 s4ener:RemainingPauseTime	The duration that the current slot permits being paused. This element SHALL ONLY be present if the power sequence is interruptible (pausable), i.e. saref:isInterrupionPossible has value TRUE.
NOTE 1: This element applies to every repetition of the slot number. NOTE 2: This element applies to the first repetition of the slot number only. NOTE 3: This element applies to the first repetition of the slot number only.	

4.2.6 Load control

This clause presents the part of SAREF4ENER that defines how to model events used in, for example, a direct load management and power curtailing scenarios (i.e. use case 4). The classes of interest are `s4ener:LoadControlEventData`, `s4ener:LoadControlEventAction`, `s4ener:LoadControlStateData` and `s4ener:LoadControlState`, as shown in Figure 7.

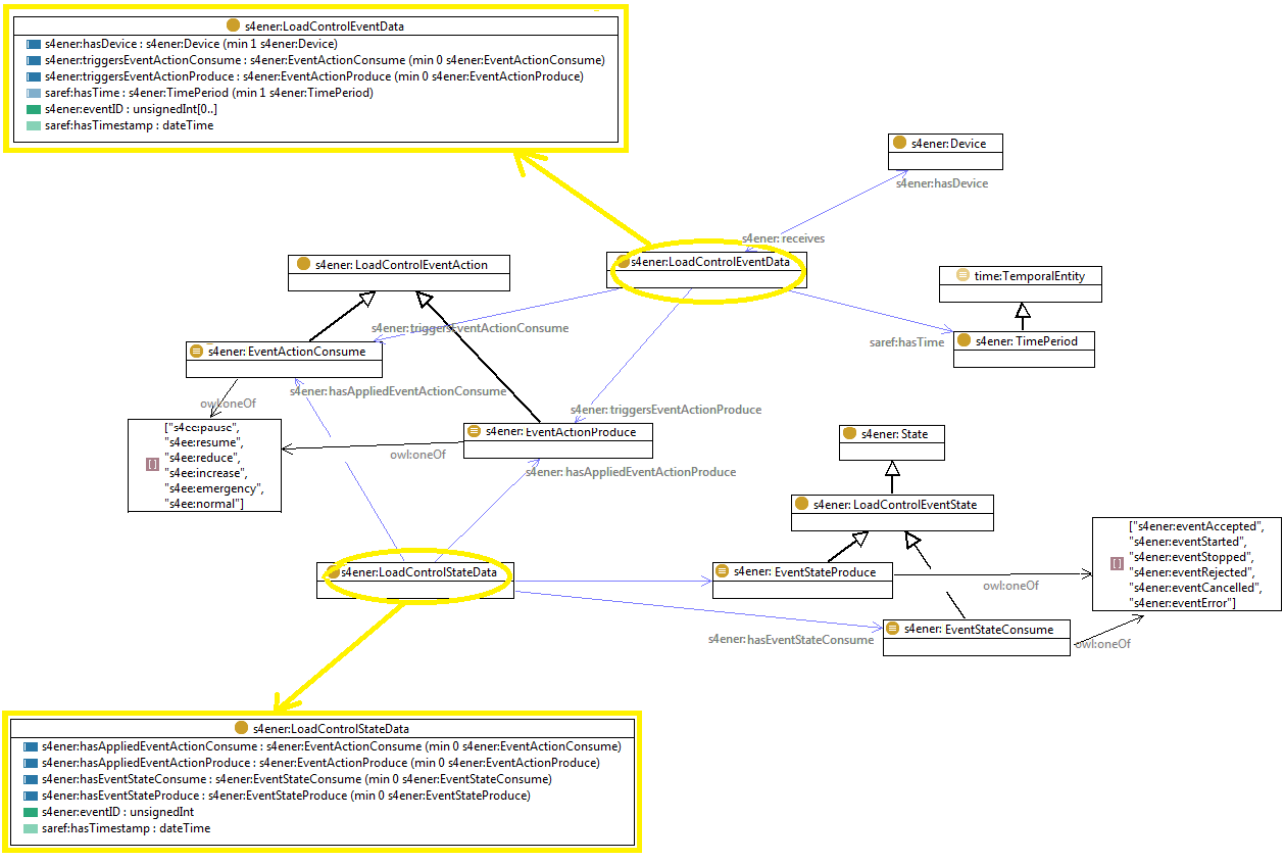


Figure 7: Load Control

The `s4ener: LoadControlEventData` class is used to represent overload warning severity level and related load control commands to a device. It is characterized by an event ID and a timestamp that represents the time the event information instance was created or received, and the time period that denotes the period of validity of the event. For example, 5 minutes ago an event was received which says that it shall take effect tomorrow from 14:00 to 15:30. In this event the timestamp is "5 minutes ago" and time period is "tomorrow from 14:00 to 15:30".

The `s4ener:LoadControlEventAction` class expresses the type of actions to be performed as a consequence of a load control event. A `s4ener:LoadControlEventAction` can be of type "consume" or "produce" to denote consumption or production of energy or power. Values for both consume and produce actions can be `s4ener:emergency`, `s4ener:increase`, `s4ener:normal`, `s4ener:pause`, `s4ener:reduce`, `s4ener:resume`.

The `s4ener: LoadControlStateData` class expresses the data about the state of an event and is characterized by the same event ID used in the `s4ener:LoadControlEventData` class, as well as a timestamp, and it is associated to the class `s4ener:LoadControlState`, which can be of type "consume" or "produce" - analogously to a load control event action – and expresses the possible states of a load control event. Values for both consume and produce load control states can be `s4ener:eventAccepted`, `s4ener:eventStarted`, `s4ener:eventStopped`, `s4ener:eventRejected`, `s4ener:eventCancelled`, or `s4ener:eventError`.

4.3 Observations about SAREF4ENER

The extension for the energy domain presented in the present document was originally called SAREF4EE, since it was created for the Energy@Home and EEBus associations [i.1]. However, in the present document the extension has been renamed to SAREF4ENER according to the naming convention for SAREF extensions adopted in the ETSI TR 103 411 [i.4] (i.e. SAREF4XXXX, where XXXX are letters that describe the domain for which the extension was created).

The present document describes the concepts for the use case 2 (smart energy management), and the use case 4 (representation of events in case of direct load management and power curtailing) elaborated in clause 5.1.1 in ETSI TR 103 411 [i.4]. However, the present document does not include the concepts for the use case 1 (exchange configuration information of devices in order to connect to each other) and the use case 3 (monitor and control the start and status of the appliances), since these concepts at the time of publication are under discussion between the Energy@Home and EEBus associations and NOT yet included in the SPINE specification in [1]. For the sake of completeness, these concepts are included in the annex B of the present document as informative, but they are NOT part of the current release of SAREF4ENER.

Annex A (informative): Approach

The approach that was followed to create SAREF4ENER is a combination of bottom-up and top-down steps, as shown in Figure A.1. The (bottom-up) starting point is given by the two existing data models of E@H (an UML class diagram) and EEBus (the XSDs specification). These two data models focus on similar concepts, such as the concept of "power profile", but they use different terminologies. For example, E@H defines "power profiles", "modes" and "phases", while EEBus refers to these concepts as "power sequences", "alternatives" and "slots". In order to converge to a shared terminology, experts of EEBus and E@H preliminarily defined a common specification [1] that was subsequently used by TNO as basis for creating SAREF4ENER.

The preliminary phase was followed by a kick-off workshop in which the experts of EEBus and E@H presented the details of their individual data models, i.e. EEBus (XSDs) and E@H (UML), and also their common data model, the EEBus & E@H (UML+XSDs) model.

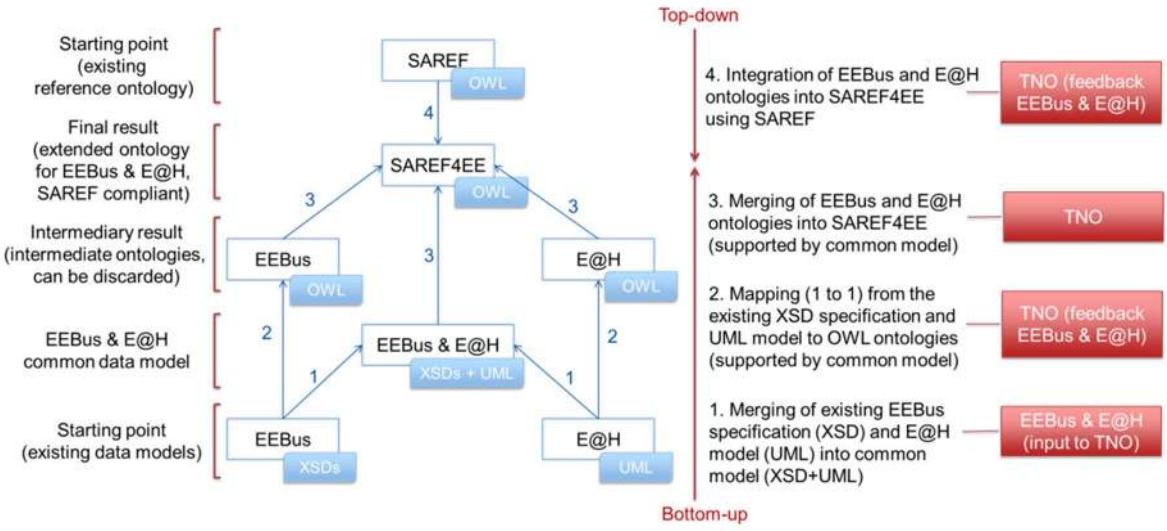


Figure A.1: Approach

Since the existing EEBus and E@H data models were expressed in different formats, i.e. XSD and UML, and SAREF4ENER had to be expressed in OWL as an extension of SAREF, these data models were first translated into corresponding OWL versions that could be used as intermediate ontologies towards the creation of SAREF4ENER. The transformations UML → OWL and XSD → OWL were performed manually, but existing tools can be used to automate this step (for example, TopBraid Composer™ Maestro Edition). The outcomes of these transformations were the EEBus (OWL) and E@H (OWL) intermediate ontologies in Figure A.1. The reason to create these two separate intermediate ontologies was practical. The common EEBus & E@H data model is a merged model whose parts could be straightforwardly identified as coming either from the EEBus or the E@H data model. Given that the EEBus and E@H experts were not yet (completely) acquainted with ontologies and OWL, their review process was facilitated by separating the generation of an OWL version in two parts. In this way, these experts could focus on their own part, namely EEBus or E@H, instead of having to deal with a single, large and more complex ontology. Moreover, these intermediate ontologies can be reused individually by the two associations if they decide to make use of an OWL version of their own data model in the future.

NOTE: TopBraid Composer Maestro Edition™ is an example of a suitable product available commercially. This information is given for the convenience of users of the present document and does not constitute an endorsement by ETSI of this product.

After receiving and incorporating the feedback from EEBus and E@H experts, the two intermediate ontologies were merged into a first version of SAREF4ENER, as depicted by step 3 in Figure A.1. Since this initial SAREF4ENER version was obtained by making a one-to-one mapping of existing data models that were implementation-driven rather than conceptual specifications, it was necessary to:

1) cleanse unnecessary redundancy, e.g. redundancy of data type properties carrying the same semantics, especially when expressing time-related information and unit of measures; and

2) create axioms that were absent in the original data models. While doing so, a top-down approach starting from SAREF was taken, as depicted by step 4 in Figure A.1. SAREF contains concepts that are rather high-level and needed further specialization into a finer-grained level of detail to accommodate the specific requirements of the EEBus and E@H use cases.

Therefore, when creating SAREF4ENER, classes and properties of SAREF were reused and specialized where possible, while SAREF was extended with new classes and properties where it did not suffice for the purpose.

In particular:

- Only a subset of concepts defined in SAREF was reused, i.e. saref:Device, saref:Profile, saref:State, saref:Energy, saref:Power, saref:UnitOfMeasure and saref:Time.

- The saref:Device and saref:Profile classes were specialized in the more specific s4ener:Device and s4ener:PowerProfile subclasses, respectively. Devices and power profiles in SAREF4ENER have specific properties for EEbus and E@H that do not apply to all SAREF devices and profiles.

Annex B (informative): Additional concepts

This annex presents some additional concepts concerning how devices can exchange configuration information on their mutual functionality in order to connect to each other (i.e. use case 1), and how the start and status of the appliances can be monitored and controlled (i.e. use case 3). These use cases are elaborated in clause 5.1.1 in ETSI TR 103 411 [i.4]. Note that the concepts described in the present annex are NOT part of the current release of SAREF4ENER, since at the time of publication of the present document they are under discussion between the Energy@Home and EEBus associations, and NOT yet included in their common specification in [1]. Note also that the concepts shown in this annex still use the prefix `s4ee:` because they are part of the first extension created for the Energy@Home and EEBus associations that was called SAREF4EE. These SAREF4EE concepts are not part yet of the extension that is now called SAREF4ENER, therefore they keep the `s4ee:`prefix.

The classes of interest for the use case 1 are `s4ee:Device`, `s4ee:Address`, `s4eeDeviceConnection`, `s4eeDeviceConnectionSetup`, `s4eeNativeSetup`, `s4eeCandidateSetup`, `s4eeScanSetup` and `s4eeJoinModeConfiguration`, as shown in Figure B.1.

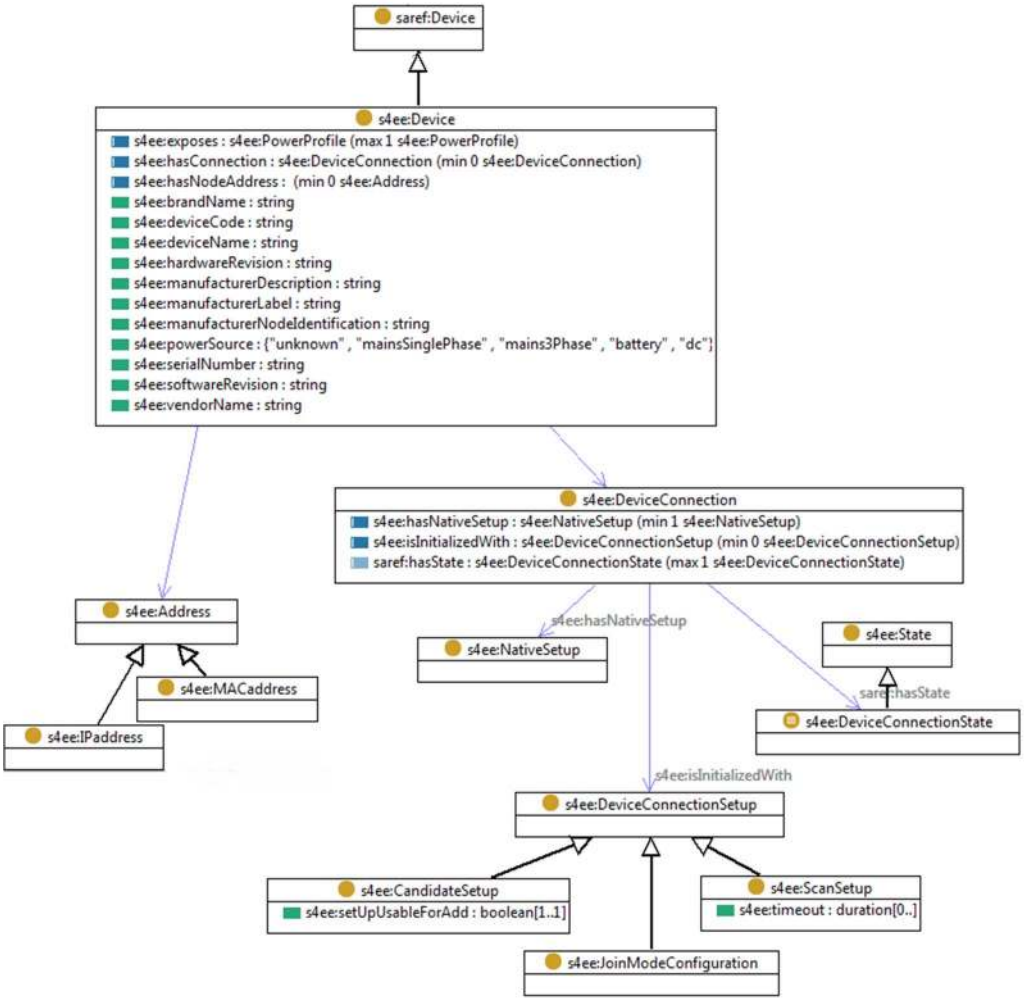


Figure B.1: Configuration

The classes of interest for the use case 2 are `s4eeAppliance`, `s4eeApplianceParameter`, `s4eeApplianceParameterTable`, `s4eeParameterTablePoint`, `s4eeValue`, `s4eeApplianceParameterState`, `s4eeApplianceWorkingMode`, `s4eeApplianceParameterSet`, `s4eeApplianceParameterSettings`, `s4eeExpression`, `s4ee`ApplianceParameterCompatibilityAction, `s4ee`ApplianceControl and `s4eeApplianceMonitor`.

A `s4eeAppliance` is as a specialization of a `s4eeDevice`and therefore also a specialization of a `saref:Device`, as shown in Figure B.2.

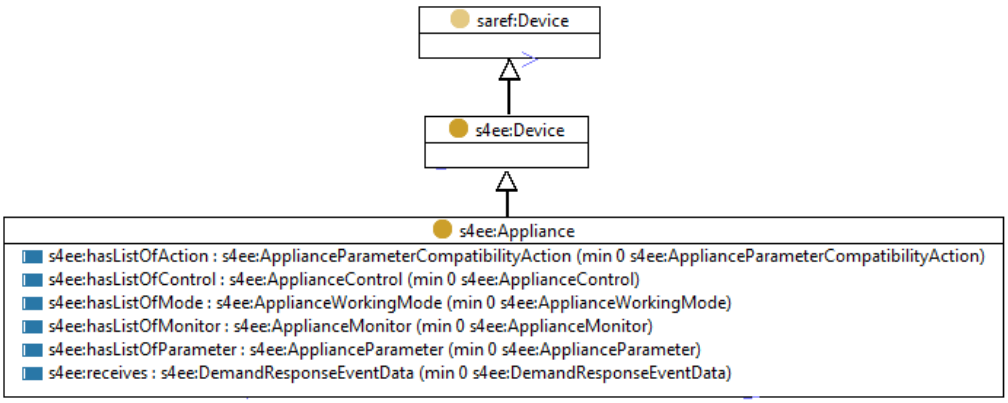


Figure B.2: Appliance

A `s4eeAppliance` is linked to parameters , available working modes, controls and measurements, as follows:

- It has a list of zero or more parameters (`s4eeApplianceParameter` class in Figure B.3), each representing a particular function mode such as "Temperature", "Spin", "Prewash" or "Iron Min":
  - Each s4eeApplianceParameter is described by s4eeParameterTable, which can be of type s4eeStepParameterTable, s4eePointwiseParameterTable, s4eeBooleanParameterTable or s4eeDateParameterTable. All these tables define the type of permission for a certain parameter (i.e. "read only", "write only" or "read and write") and its unit of measure (saref:isMeasuredIn property). The s4eeStepParameterTable is additionally characterized by at least one minimum value, number of set points and steps. The s4eePointwiseParameterTable is characterized by a point with one or more values (s4eehasPoint min 1 property) described by the s4eeParameterTablePoint class.
  - Each s4eeApplianceParameter is associated to a state (saref:hasState exactly 1 s4eeApplianceParameterState) which can be used to represent the actual parameter values by means of the s4eeApplianceMonitor class, and to set new values by means of the s4eeApplianceControl class.

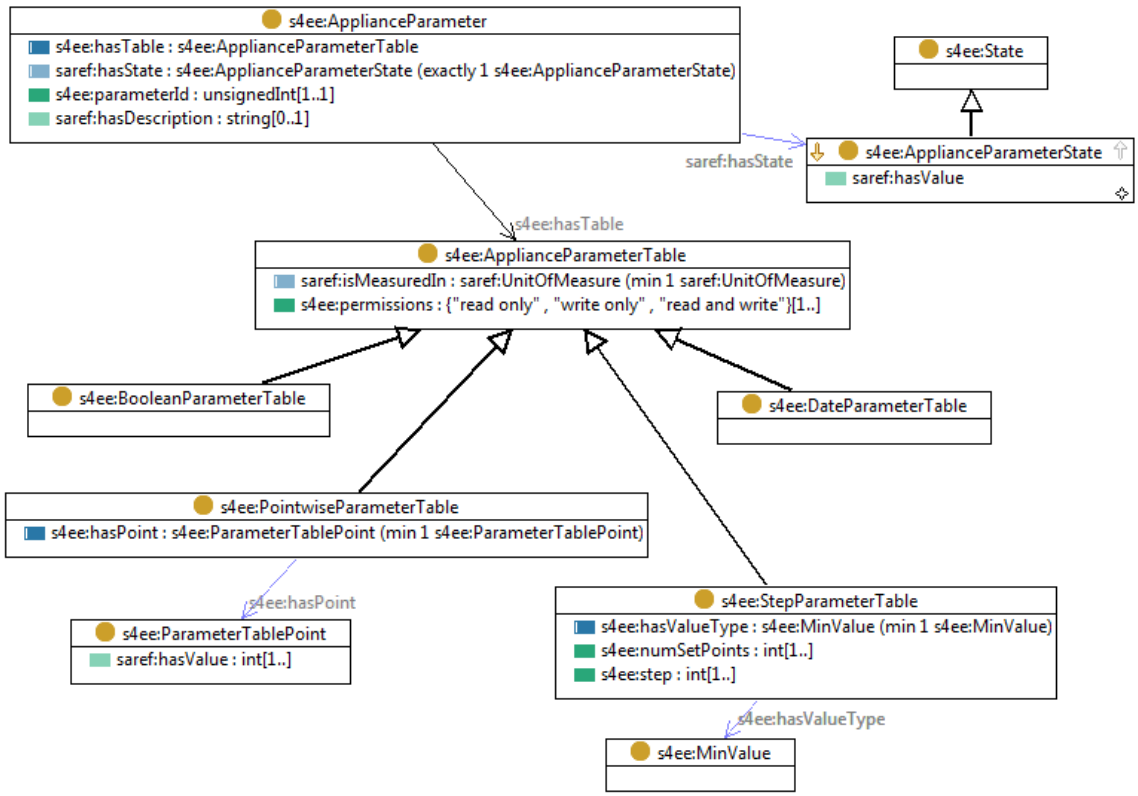


Figure B.3: ApplianceParameter

- It has a list of zero or more working modes (`s4eeApplianceWorkingMode` class in Figure B.4) each representing a particular working mode such as "Synthetics", "Mix 30" or "Super Cool":

  - A working mode has an ID (`s4eeworkingModeId exactly 1` property), a name (`saref:hasName exactly 1` property) and a list of zero or more sets (`s4eeApplianceParameterSet` class) representing the sets of enabled parameters for that working mode. A `s4ee:ApplianceParameterSet` can have zero or more settings (`s4eeApplianceParameterSettings` class)and is selected according to certain conditions defined in the `s4eeExpression` class. The set "0" is the default set and is selected when no condition is true:

    - The `s4ee:ApplianceParameterSettings` class is characterized by the parameter ID (`s4eeparameterId exactly 1` property) and a number of values for that parameter that are subclass of the `s4eeValue` class, i.e. `s4eeAvoidedValue` (list of not admitted values), `s4ee:DefaultValue` (default value of the parameter), `s4eeMaxValue` (maximum value that the parameter could be set) and `s4eeMinValue` classes (minimum value that the parameter could be set). The `s4eeApplianceParameterSettings` class also has a boolean property to specify whether the settings under consideration are active or not (`s4eeisActive` property).

    - The s4ee`:Expression` class is characterized by a value (`s4eehasValueType exactly 1` property), the parameter ID (`s4ee`parameterId exactly 1 property) that identifies the parameter whose current set point has to be compared, a math operator `s4eemathOperator exactly 1` property) such as "==" , "!=" , ">" , "<" to define set points equal, different, above or below the expression value, and logical connectives (`s4eelogicalConnective min 0` property) such as "AND" and "OR" that could be used to connect different expressions.

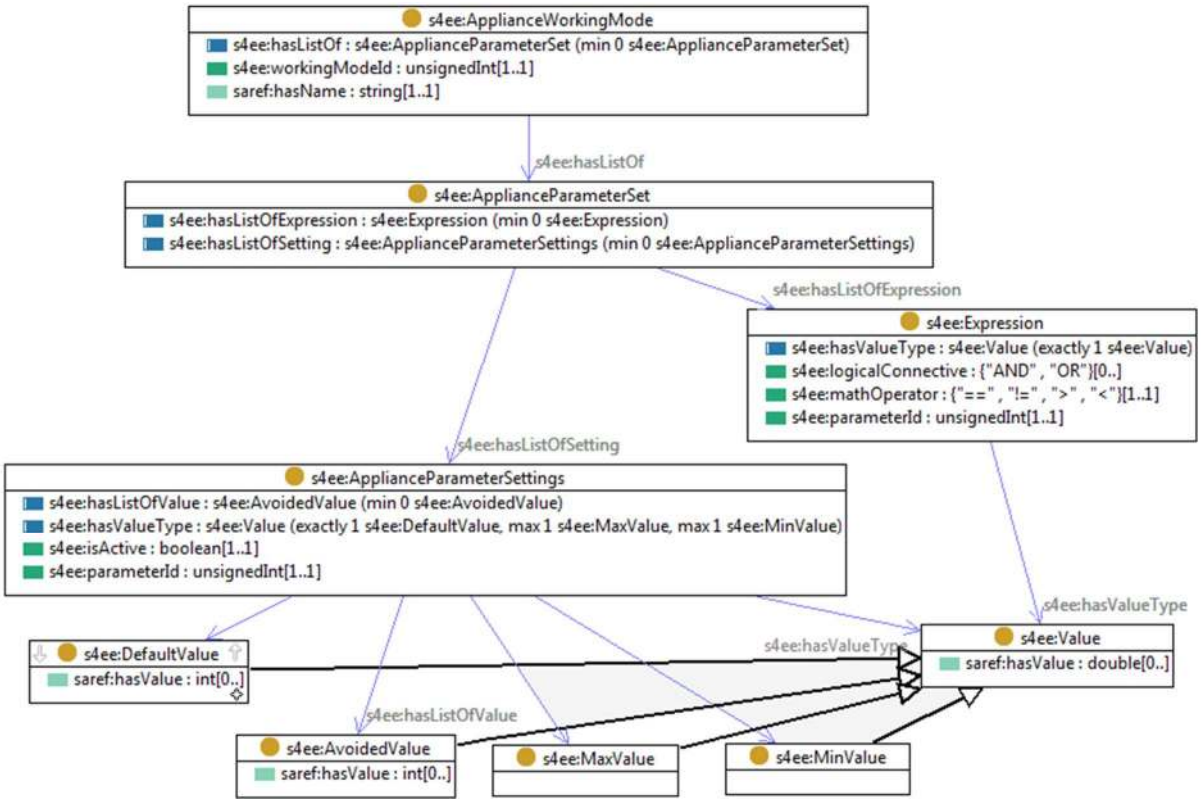


Figure B.4: ApplianceWorkingMode

- It has a list of zero or more actions to be executed in case of incompatibility with other parameters (`s4ee:ApplianceParameterCompatibilityAction` class in Figure B.5):

  - The `s4eeApplianceParameterCompatibilityAction` class specifies incompatible parameters (`s4eehasAffectedParameter min 1` property), and has at least one expression (`s4eehasExpression min 1` property). If this expression turns TRUE, then one of the following types of actions will be executed:

    - s4eeaction_1_reset_to_OFF_value (reset).
    - s4eeaction_2_disabled (disabled).
    - s4eeaction_3_set_to_MaxValue (set to maximum value).
    - s4eeaction_4_set_to_default_value (set to default value).

The property `s4eehasValue min 0 s4eeMaxValue` expresses the maximum value to be used in case of `s4eeaction_3_set_to_MaxValue`.

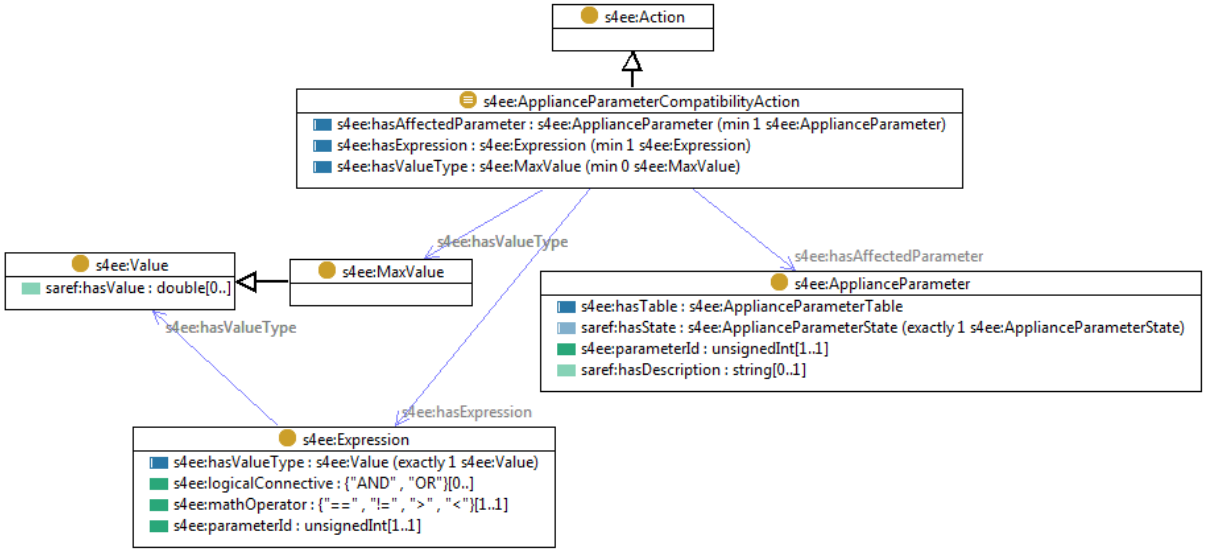


Figure B.5: ApplianceCompatibilityAction

- It has a list of zero or more measurements that represent the actual parameter values for the appliance (`s4eeApplianceMonitor` class in Figure B.6). These measurements can be sent by the appliance automatically as a status notification, or after a specific request from the CEM. The notification contains the information related to the current state of the appliance, i.e. parameter ID, its current value and, optionally, the maximum and minimum values that the parameter can assume.

- It has a list of zero or more control actions (`s4eeApplianceControl` class in Figure B.6), such as command actuation or the setting of working modes and parameters, to control zero or more states of the appliance (`s4eeApplianceParameterState` class). The `s4eeApplianceControl` class also has a boolean property to specify whether the controls under consideration are active or not (`s4eeisActiveyioy` property).

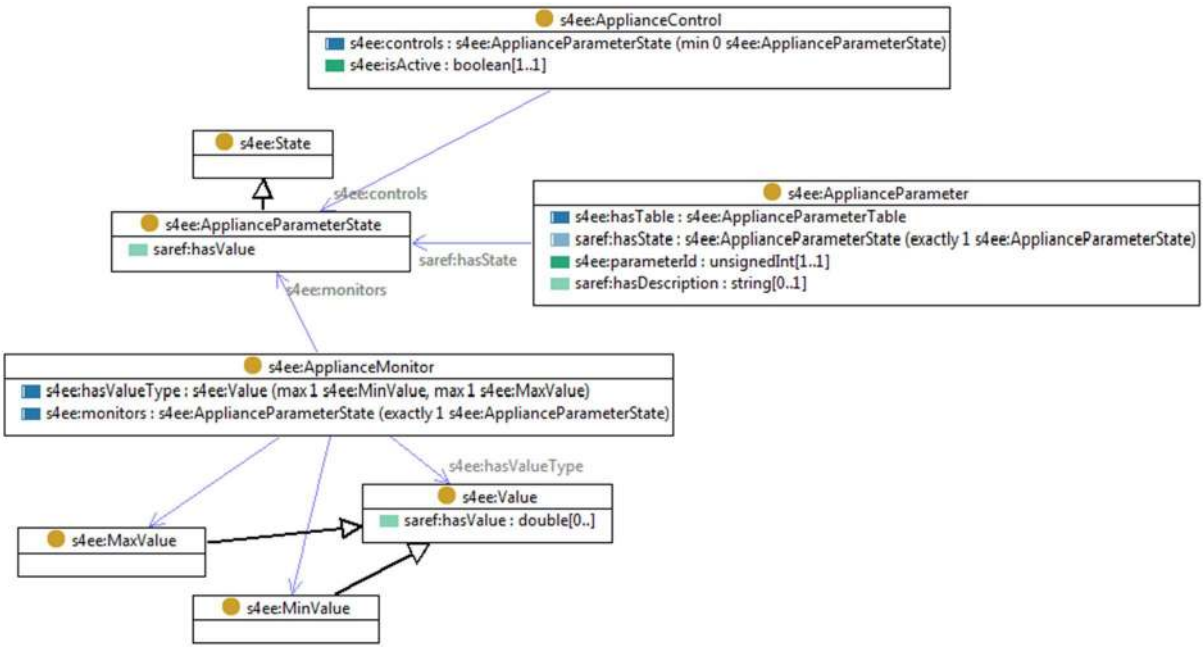


Figure B.6: ApplianceMonitor and ApplianceControl

Annex C (informative): Bibliography

- ETSI TS 103 267: "SmartM2M; Smart Applications; Communication Framework".
- ETSI TS 102 689: "Machine-to-Machine communications (M2M); M2M Service Requirements".
- ETSI TS 118 101: "oneM2M; Functional Architecture (oneM2M TS-0001)".
- ETSI TS 118 102: "oneM2M; Requirements (oneM2M TS-0002)".
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NOTE: Available at https://sites.google.com/site/smartappliancesproject/deliverables.

History

Document history		
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